

IT-Simplera Solutions



Week 3 Task Report

Enterprise Multi-Protocol Routing Challenge (OSPF Multi-Area, Static Routing & RIP v1/v2)

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Date Of Submission

18th July, 2026

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1. Network Design & Core Concepts

1.1. Open Shortest Path First (OSPF) Multi-Area

- **Definition:** OSPF is a link-state routing protocol that uses the Shortest Path First (SPF) algorithm to determine the best path to a destination. Multi-Area OSPF divides a large routing domain into smaller logical groups (areas) to localize routing updates and reduce SPF recalculation overhead.
- **Explanation:** By restricting topology changes to a specific area (Area 1 or Area 2) and requiring all non-backbone areas to connect through a central backbone (Area 0), the network reduces the size of each router's link-state database and the CPU cost of recomputing routes.

1.2. Routing Information Protocol (RIP v1/v2)

- **Definition:** RIP is a distance-vector routing protocol that uses hop count as its metric to determine the best path, with a maximum of 15 hops.
- **Explanation:** RIPv1 is a classful protocol and does not transmit subnet mask information in its updates, meaning it cannot support Variable Length Subnet Masking (VLSM). RIPv2 addresses this by carrying the subnet mask with every route, supporting classless (VLSM) addressing, and using multicast (224.0.0.9) instead of broadcast.

1.3. Route Redistribution

- **Definition:** The process of taking routing information learned by one routing protocol (e.g. RIP) and injecting it into another routing protocol (e.g. OSPF) so that both domains share reachability information.
- **Explanation:** Because different protocols use incompatible metrics (OSPF uses cost; RIP uses hop count), redistribution requires an explicit seed metric on the boundary router so the receiving protocol knows how to rank the imported routes.

1.4. Floating Static Routes

- **Definition:** A backup static route configured with a higher Administrative Distance (AD) than the primary route to the same destination.
- **Explanation:** The routing table only installs the route with the lowest AD. If the primary path fails, the floating static route becomes the best (and only) available path and is automatically installed, restoring connectivity without manual intervention.

2. Configuration & Verification Guide

The network topology below shows all seven routers grouped by routing domain: two OSPF non-backbone areas (Area 1, Area 2), the OSPF backbone (Area 0), the RIP redistribution point (R4, acting as ASBR), the RIPv2 and RIPv1 sites (R5, R6), and the static-routed remote office (R7) with a primary and floating backup path.

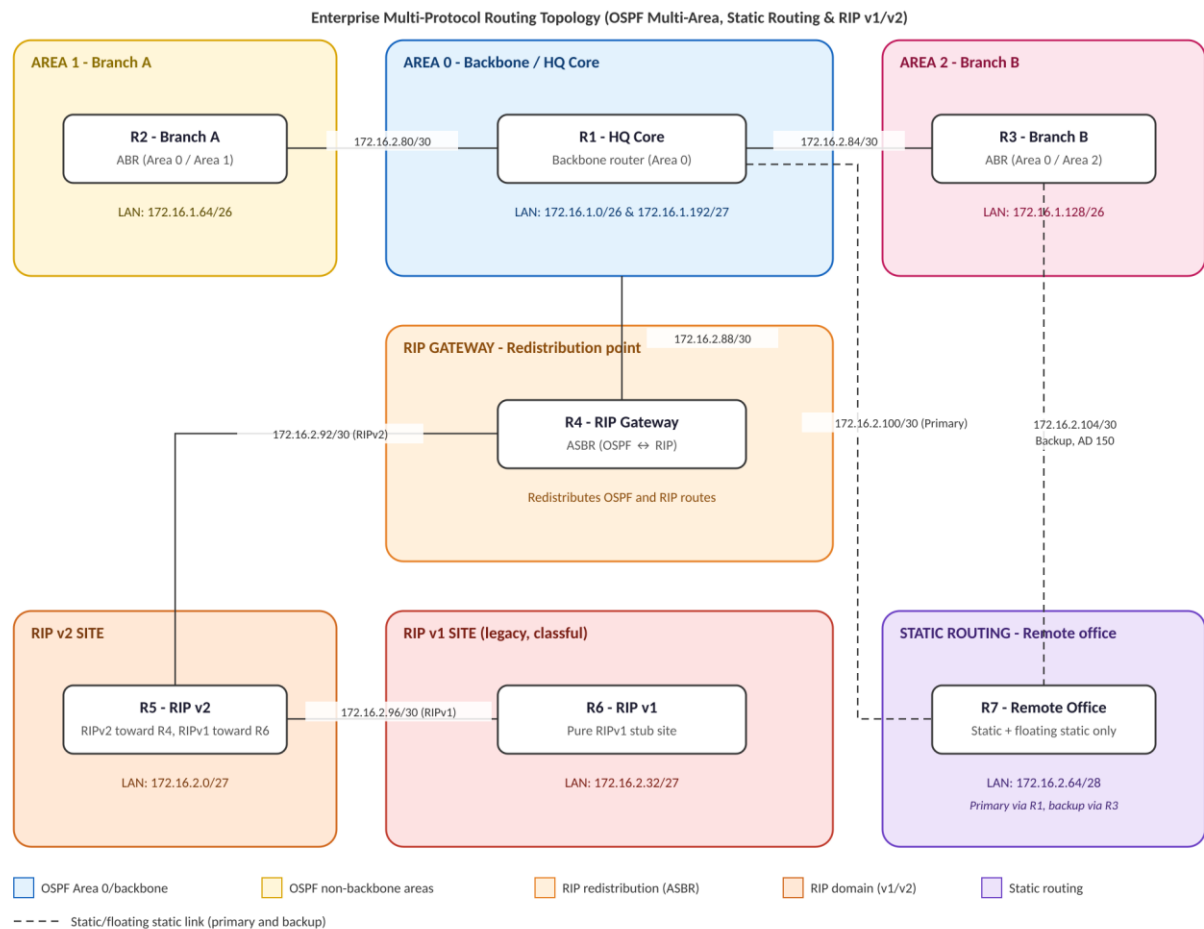
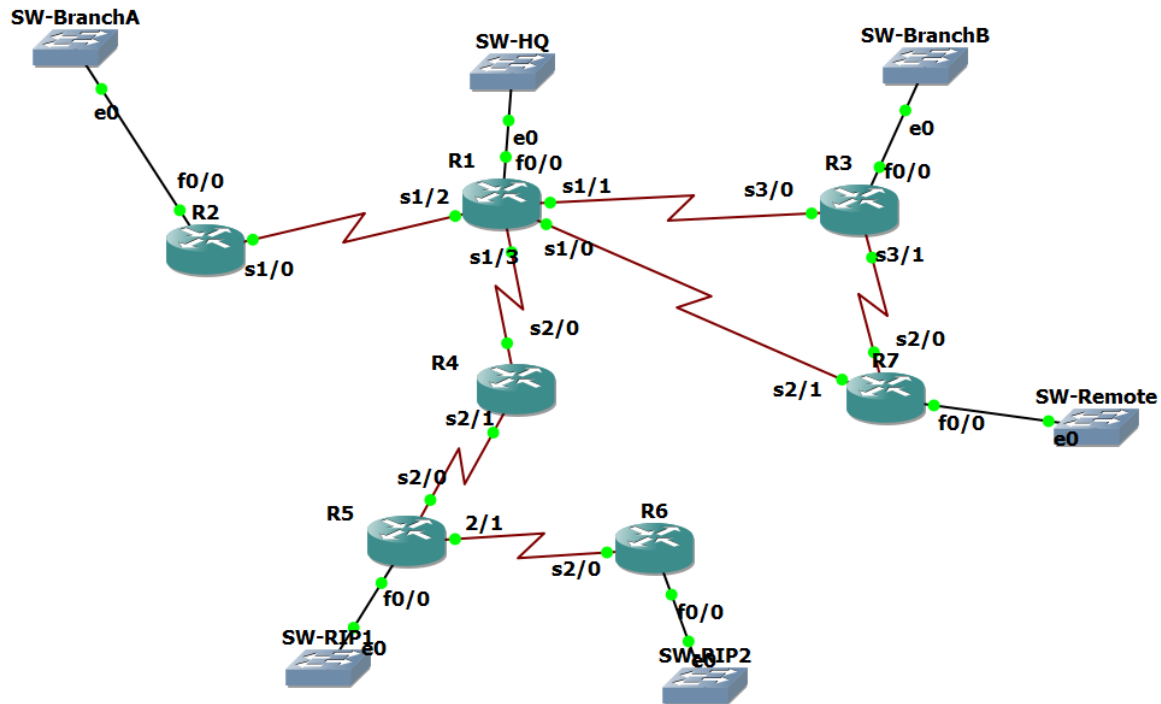


Figure 1: Enterprise multi-protocol routing topology, grouped by routing domain.



2.1. IP Addressing Table (VLSM)

LAN subnets

#	Network / Subnet	Usable Range	Used By
1	172.16.1.0/26	.1 - .62	R1 - HQ Admin VLAN
2	172.16.1.192/27	.193 - .222	R1 - HQ Server VLAN
3	172.16.1.64/26	.65 - .126	R2 - Branch A LAN
4	172.16.1.128/26	.129 - .190	R3 - Branch B LAN
5	172.16.2.0/27	.1 - .30	R5 - RIPv2 Site LAN
6	172.16.2.32/27	.33 - .62	R6 - RIPv1 Site LAN
7	172.16.2.64/28	.65 - .78	R7 - Remote Office LAN

WAN links (point-to-point /30)

#	Network / Subnet	Router A	Router B
8	172.16.2.80/30	R1 (.81)	R2 (.82)
9	172.16.2.84/30	R1 (.85)	R3 (.86)
10	172.16.2.88/30	R1 (.89)	R4 (.90)
11	172.16.2.92/30	R4 (.93)	R5 (.94) - RIPv2 side
12	172.16.2.96/30	R5 (.97)	R6 (.98) - RIPv1 side
13	172.16.2.100/30	R1 (.101)	R7 (.102) - Primary
14	172.16.2.104/30	R3 (.105)	R7 (.106) - Backup, AD 150

2.2. OSPF Neighbor Verification

To confirm that routers are exchanging topology data correctly within the multi-area design:

- **Command:** show ip ospf neighbor
- **Explanation:** This verifies that neighbors have reached the FULL state, indicating successful link-state database synchronization across the backbone and each non-backbone area.

OSPF Neighbor Output

Router 1 (backbone core - three adjacencies across Area 0):

```
R1#show ip ospf neighbor
Neighbor ID  Pri  State      Dead Time  Address      Interface
172.16.255.4  0  FULL/-    00:00:33  172.16.2.90  Serial1/2
172.16.255.3  0  FULL/-    00:00:34  172.16.2.86  Serial1/1
172.16.255.2  0  FULL/-    00:00:38  172.16.2.82  Serial1/0
```

Router 4 (RIP gateway / ASBR - adjacency with the backbone):

```
R4#show ip ospf neighbor
Neighbor ID  Pri  State      Dead Time  Address      Interface
172.16.255.1  0  FULL/-    00:00:33  172.16.2.89  Serial2/0
```

Note: neighbor verification tables for R2 and R3 are included in the screenshots folder.

2.3. Routing Table & Redistribution Verification

To confirm that OSPF, RIP, and static routes are correctly redistributing across the network:

- **Command:** show ip route ospf / show ip route rip
- **Explanation:** Check for codes such as O IA (OSPF inter-area), O E2 (OSPF external, redistributed from RIP or static), and R (native RIP) to confirm that routes are crossing domain boundaries correctly.

Redistribution snapshot - Router 1

```
R1#show ip route ospf
172.16.0.0/16 is variably subnetted, 21 subnets, 4 masks
O IA  172.16.1.64/26 [110/65] via 172.16.2.82, Serial1/0
O IA  172.16.1.128/26 [110/65] via 172.16.2.86, Serial1/1
O E2  172.16.2.0/27 [110/20] via 172.16.2.90, Serial1/2
O E2  172.16.2.92/30 [110/20] via 172.16.2.90, Serial1/2
O E2  172.16.2.96/30 [110/20] via 172.16.2.90, Serial1/2
O IA  172.16.255.2/32 [110/65] via 172.16.2.82, Serial1/0
O IA  172.16.255.3/32 [110/65] via 172.16.2.86, Serial1/1
O    172.16.255.4/32 [110/65] via 172.16.2.90, Serial1/2
O E2  172.16.255.5/32 [110/20] via 172.16.2.90, Serial1/2
O E2  172.16.255.6/32 [110/20] via 172.16.2.90, Serial1/2
```

The O IA entries confirm inter-area OSPF routing (Area 1 and Area 2 reachable from the backbone). The O E2 entries confirm RIP routes successfully redistributed into OSPF via R4.

Redistribution snapshot - Router 4 (RIP side)

```
R4#show ip route rip
  172.16.0.0/16 is variably subnetted, 17 subnets, 4 masks
R   172.16.2.0/27 [120/1] via 172.16.2.94, Serial2/1
R   172.16.2.96/30 [120/1] via 172.16.2.94, Serial2/1
R   172.16.255.5/32 [120/1] via 172.16.2.94, Serial2/1
R   172.16.255.6/32 [120/2] via 172.16.2.94, Serial2/1
```

Note: full routing table verification for R2, R3, R5, R6 and R7 is included in the screenshots folder.

2.4. Ping Test and Traceroute Results

Connectivity was verified across every routing domain boundary in the topology: OSPF inter-area, OSPF to RIP (both directions), RIP to static, and OSPF to static.

Cross-domain reachability (OSPF to Static, longest path in the network)

```
R2#ping 172.16.2.65
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.2.65, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 36/59/88 ms

R2#traceroute 172.16.2.65
Type escape sequence to abort.
Tracing the route to 172.16.2.65
 1 172.16.2.81 52 msec 24 msec 28 msec
 2 172.16.2.102 20 msec 60 msec 60 msec
```

Traceroute confirms the packet crosses OSPF (R2 to R1) and then the static route on R1 to reach R7 - exactly two hops, matching the design.

RIP (v1 stub site) to OSPF backbone

```
R6#ping 172.16.1.1
Success rate is 100 percent (5/5), round-trip min/avg/max = 96/128/140 ms
```

RIP domain to Static domain (crosses all three routing protocols)

```
R6#ping 172.16.2.65
Success rate is 100 percent (5/5), round-trip min/avg/max = 176/180/188 ms
```

OSPF backbone to RIPv1 stub LAN

```
R1#ping 172.16.2.33
Success rate is 100 percent (5/5), round-trip min/avg/max = 88/106/140 ms
```

Note: the complete ping matrix, covering every router pair, is included in the screenshots folder.

3. Troubleshooting Report

3.1. Issue Description

- **Scenario:** Testing static route redundancy and failover for the remote office (R7), which is reached via a primary static route from R1 and a floating static backup route from R3.
- **Symptom:** When the primary link between R1 and R7 was intentionally shut down to test the backup path, the floating static route needed to activate automatically and traffic needed to reroute through R3 without manual reconfiguration.

3.2. Root Cause Analysis

This was a deliberate failure simulation rather than a misconfiguration. Prior to the test, R3's floating static route (Administrative Distance 150) was correctly present in the configuration but not installed in the routing table, since R1's primary static route (AD 1) was preferred while the primary link was healthy. Shutting down R1's Serial1/3 interface removed the OSPF-redistributed route to R7's LAN, at which point R3's own floating static route became the best - and only - available path.

3.3. Resolution Steps

- Captured the baseline state on R3 (show ip route static) confirming the floating route was dormant while the primary link was healthy.
- Shut down R1's Serial1/3 interface (the primary link to R7) to simulate a link failure.
- Re-checked R3's routing table to confirm the floating static route activated automatically, and re-ran ping/traceroute from R2 to confirm traffic rerouted through R3.
- Restored the primary link (no shutdown on R1's Serial1/3) and confirmed the network reverted to the primary path once OSPF reconverged.

3.4. Verification

Before failure - floating static is dormant on R3:

```
R3#show ip route static
(no output - floating static not installed while primary route is preferred)
```

During failure - floating static becomes active:

```
R3#show ip route static
S    172.16.2.64/28 [150/0] via 172.16.2.106
```

During failure - traceroute confirms traffic now transits R3 instead of R1:

```
R2#traceroute 172.16.2.65
1 172.16.2.81 72 msec 28 msec 24 msec
```



```
2 172.16.2.86 28 msec 56 msec 60 msec
3 172.16.2.106 96 msec 80 msec 96 msec
```

The path grew from 2 hops (via R1) to 3 hops (via R3), confirming the floating static backup successfully took over. A brief packet-loss window (2 of 5 pings) was observed immediately after the link failure due to normal OSPF/routing-table convergence delay; once converged, connectivity returned to 100 percent.

After recovery - floating static deactivates and the primary path is restored:

```
R3#show ip route static
(no output - floating static deactivated, primary route preferred again)
```

Note: full before / during / after screenshots for this failover sequence are included in the screenshots folder.

4. Challenges Faced

4.1. Duplicate OSPF Router ID Between R1 and R2

- **Challenge:** After applying the initial OSPF configuration to all routers, R1 showed only one OSPF neighbor (R3) instead of the expected three. R2's Serial link to R1 was up at Layer 3 but never formed an adjacency, and R2's console displayed the log message %OSPF-4-DUP_RTRID_NBR: OSPF detected duplicate router-id 172.16.255.1 from 172.16.2.81 on interface Serial1/0.
- **Root Cause:** R2's router-id had been mistakenly copy-pasted from R1's configuration (172.16.255.1) instead of being set to its own value (172.16.255.2). Two routers cannot form an OSPF adjacency while sharing the same router-id.
- **Resolution:** Corrected R2's router-id to 172.16.255.2 under router ospf 1, then ran clear ip ospf process to force the OSPF process to restart cleanly with the new ID. The adjacency formed immediately afterward, confirmed via show ip ospf neighbor on both routers.

4.2. RIPv1 Classful Boundary Dropped a VLSM Subnet

- **Challenge:** After bringing up the RIP domain, R4's routing table showed R6's loopback address but not R6's actual LAN subnet (172.16.2.32/27), even though both were advertised by the same router over the same RIPv1 link.
- **Root Cause:** RIPv1 does not transmit subnet mask information in its updates. Cisco IOS applies a rule that a subnet is only advertised out an interface if its mask matches that interface's own mask, or is dropped as classfully ambiguous otherwise. Since R6's LAN mask (/27) did not match the outgoing serial interface's mask (/30), the route was silently withheld - a textbook demonstration of why RIPv1 does not support VLSM.
- **Resolution:** Configured R5 (the RIPv1/RIPv2 boundary router) to originate a default route into RIP for the stub site (default-information originate) and to explicitly redistribute a static route for R6's LAN over its RIPv2-facing link toward R4, where subnet masks are fully supported. This restored bidirectional

reachability to R6's LAN without compromising the deliberate RIPv1/RIPv2 coexistence design.

5. Conclusion

This implementation successfully demonstrates multi-protocol integration across OSPF Multi-Area routing, RIP versions 1 and 2, and static routing with floating-static redundancy. Through deliberate route redistribution at R1, R3, and R4, and the deployment of a primary/backup static path to the remote office, the network achieved verified end-to-end connectivity across every routing domain boundary, including the longest path in the topology (RIP domain through OSPF to the static-routed remote office).

The troubleshooting exercises - a duplicate OSPF router-id and a genuine RIPv1/VLSM incompatibility - reinforced the importance of verifying not just that a protocol is configured, but that its adjacencies and advertised routes match expectations at every hop. The static route failover test further confirmed that a floating static route, once correctly configured with a higher administrative distance, restores connectivity automatically during a link failure without requiring manual intervention, which is the core objective of route redundancy in an enterprise network.